

## 16 Vector Calculus

**Prerequisite information for Chapter 16:** Give the *COMPLETE*—both parts—statement of the Fundamental Theorem of Calculus in One Dimension (found in your text, section 5.3).

Let  $f$  be continuous on  $[a, b]$ . Define  $g$  as the function

$$g(x) = \int_a^x f(t) dt \text{ is continuous on } [a, b], \text{ differentiable on } (a, b) \text{ and } g'(x) = f(x)$$

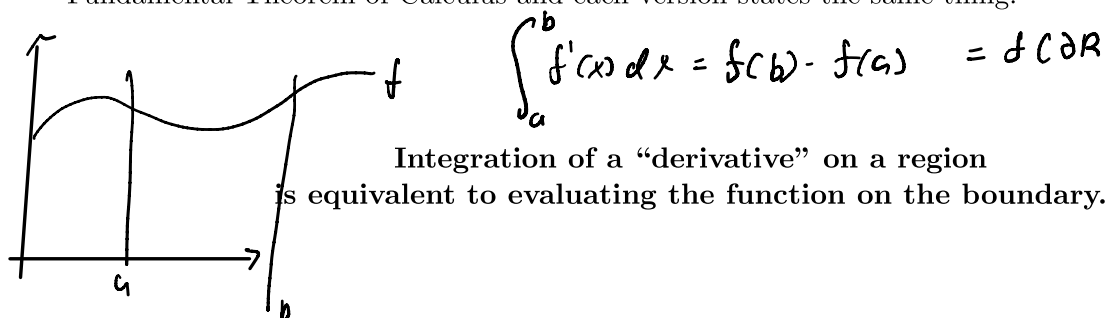
$$\text{Let } F'(x) = f(x) \text{ with } f \text{ continuous on } [a, b], \text{ then } \int_a^b f(x) dx = F(b) - F(a)$$

What parallels do you notice between this theorem and the statement, *Integration of a derivative on a region is equivalent to evaluating the function on the boundary*?

$$\frac{d}{dx} \int_a^b f(x) dx = \frac{d}{dx} (F(b) - F(a)) = \underline{f(b) - f(a)}$$

evaluating function on boundary

**An Overview of Chapter 16:** At its heart, what does the Fundamental Theorem of Calculus (from Calc 1) say? Roughly, it says that when you integrate a function's derivative on an interval, it's the same as evaluating that function on the *boundary* of that interval. (You may not have thought of the endpoints of an interval as its boundary before. Try it out. We often use the symbol,  $\partial$  to mean “boundary of.”) Chapter 16 covers the multi-variate versions (there are four) of the Fundamental Theorem of Calculus and each version states the same thing:



The term “derivative” is in quotes because multivariate functions and vector fields have three different well-studied types of derivatives. We will need to come up with a version of the Fundamental Theorem for each. (One of these derivative types gives rise to *two* versions of the theorem—one for 2-D and one for 3-D. That’s why there are four versions of the theorem.)

Before we talk about anything in chapter 16, I want to show you all **FIVE** versions of the Fundamental Theorem of Calculus (including the 1-D version you already know), so that you can see right away the commonalities in their formats. There will be notation and terminology that you do not yet know—don’t worry about that. Just look at the forms and familiarize yourself with the common pattern.

## 16.1 Vector Fields

**Definitions in this section:** Give the definition of each term below, using the given passage from your text/e-text.

- **Vector Field on  $\mathbb{R}^2$ :** [See definition in Box # 1 of Section 16.1.]

Let  $D \subseteq \mathbb{R}^2$ . A vector field on  $\mathbb{R}^2$  is a function  $\vec{F}$  that maps each  $(x, y) \in D$  to some  $\vec{F}(x, y)$

$$\vec{F}: D \rightarrow \mathbb{R}^2$$

- **Vector Field on  $\mathbb{R}^3$ :** [See definition in Box # 2 of Section 16.1]

Same defn as above except  $E \subseteq \mathbb{R}^3$  and

$$\vec{F}: E \rightarrow \mathbb{R}^3$$

- **Component Functions:** [See discussion before Figure 3 and before Figure 4 in Section 16.1.]

If  $\vec{F} = P\hat{i} + Q\hat{j}$  or  $\vec{F} = P\hat{i} + Q\hat{j} + R\hat{k}$ , where  $P, Q, R$  are scalar valued functions, then  $P, Q$ , (and  $R$ ) are the component functions of  $\vec{F}$ .

- **Gradient Vector Field:** [See discussion above Example 6 in Section 16.1.]

For some scalar valued  $f(x, y)$ ,  $\nabla f$  is really a vector field given by

$$\nabla f = f_x \hat{i} + f_y \hat{j}$$

and the same logic applies for some  $f(x, y, z)$ .

## 16.2 Line Integrals

Prerequisite information for this section:

Find the arclength of each curve below. You may wish to review section 13.3. You may also want to use a table of integrals for the antiderivative of  $\sec x$ . You can access the table of integrals in your ebook by selecting eBook within WebAssign. Then scroll down the table of contents to the "Formula Cards" link. Selecting this link will take you to the "Reference pages" which include the table of integrals.

$$1. \vec{r}(t) = \left\langle 2t, t^2, \frac{1}{3}t^3 \right\rangle, \text{ for } 1 \leq t \leq 2$$

$$\begin{aligned} \int_1^2 \sqrt{2^2 + (2t)^2 + (t^2)^2} dt &= \int_1^2 \sqrt{4 + 4t^2 + t^4} dt \\ &= \int_1^2 t^2 + 2 dt = \frac{1}{3}(2^3 - 1) + 2(2 - 1) = 2 + \frac{7}{3} \end{aligned}$$

$$2. \vec{r}(t) = \cos t \vec{i} + \sin t \vec{j} + \ln(\cos t) \vec{k} \text{ for } 0 \leq t \leq \frac{\pi}{4} \quad \frac{1}{\cos(t)} (-\sin(t)) = -\tan(t)$$

$$\begin{aligned} \int_0^{\pi/4} \sqrt{\sin^2 t + \cos^2 t + \tan^2(t)} dt &= \int_0^{\pi/4} \sqrt{1 + \tan^2 t} dt \\ &= \int_0^{\pi/4} \sec t dt \approx 0.8814 \end{aligned}$$

**Definitions in this section:** Give the definition of each term below, using the given passage from your text/e-text.

- **Piece-wise smooth curve:** [See discussion after Figure 3 in Section 16.2.]

A union of a finite number of smooth curves  $C_1, C_2, \dots, C_n$

- **Orientation of a parametrized curve:** [See discussion above Figure 8 in Section 16.2.]

A given parametrization determines an orientation of some curve  $C$   
where the positive direction is the direction of increasing  $t$ .

**Topics to learn in this section:**

1. The meaning of the line integral of a scalar function  $f(x, y)$  along a curve  $C$ .
2. The meaning of  $\int Pdx + Qdy$  along a curve  $C$ .
3. Vector fields and work: the meaning of  $\int_C \vec{F} \cdot d\vec{r}$ .

### 16.3 The Fundamental Theorem for Line Integrals

#### Prerequisite information for this section:

Consider the unit circle,  $x^2 + y^2 = 1$ , and give two parametrizations of this curve in two dimensions, as indicated below. (The first parametrization should be quite familiar; the second will be nonstandard. You may wish to review sections 13.1 through 13.3.)

1. Give a parametrization of the curve which follows the unit circle once around (counterclockwise), beginning at the point  $(1, 0)$ . (I recommend you begin with  $t = 0$ .)

$$\vec{r}(t) = \langle \cos(t), \sin(t) \rangle, \quad \text{where } 0 \leq t \leq$$

2. Give a (different) parametrization of the curve which follows the unit circle once around—clockwise, this time—beginning at the point  $(0, 1)$ . Again, please start  $t$  at  $t = 0$ .

$$\vec{w}(t) = \langle \sin(t), \cos(t) \rangle, \quad \text{where } 0 \leq t \leq$$

**Definitions in this section:** Give the definition of each term below, using the given passage from your text/e-text.

- **Path:** [See discussion above Figure 2 in Section 16.3.]

*Piecewise-smooth curves*

Is this type of **path** the same or different from the paths we used in section 14.2? Support your answer.

*Yes, we consider different paths which can be described as curves through a point to consider a limit's existence.*

- **Independence of Path (for an integral):** [See discussion below Figure 2 in Section 16.3.]

Let  $\vec{F}$  be a vector field continuous over some domain  $D$ .

Let  $C_1, C_2 \subset D$  be paths with the same initial and terminal points.

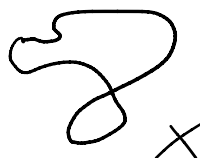
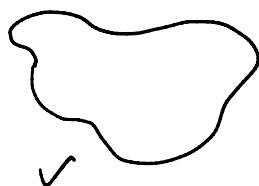
The integral  $\int_C \vec{F} \cdot d\vec{r}$  is independent of path if

$$\int_{C_1} \vec{F} \cdot d\vec{r} = \int_{C_2} \vec{F} \cdot d\vec{r}$$

- **Simply-connected Region:** [See Figure 8 and the discussion above it in Section 16.3.]

Simple curve - non intersecting

A simply-connected region is a connected region  $D$  s.t. every simple closed curve in  $D$  encloses only points in  $D$ . Intuitively, it's a region w/o holes or self-intersections.



### Topics to learn in this section:

1. The path independence of  $\int_C \nabla f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a))$  under suitable conditions.
2. The equivalence of path independence to the condition that  $\int_C \vec{F} \cdot d\vec{r} = 0$  for every closed curve  $C$  in the domain of  $\vec{F}$ .
3. The equivalence of the following three conditions on a simply-connected domain: 1) Path independence of the integral; 2)  $\vec{F}$  is conservative; 3)  $Q_x - P_y = 0$ .